This blurring is at the heart of Boltzmann's theory.[93] From this blurring, the concepts of heat and entropy are born—and these are linked to the phenomena that characterize the flow of time. The entropy of a system depends explicitly on blurring. It depends on what I *do not* register, because it depends on the number of *indistinguishable* configurations. The *same* microscopic configuration may be of high entropy with regard to one blurring and of low in relation to another.

This does not mean that blurring is a mental construct; it depends on actual, existing physical interactions.[94] Entropy is not an arbitrary quantity, nor a subjective one. It is a *relative* one, like speed.

The speed of an object is not a property of the object alone: it is a property of the object in relation to another object. The speed of a child who is running on a moving train has a value relative to the train (a few steps per second) and a different value relative to the ground (a hundred kilometers per hour). If his mother tells the child to "Keep still!," she does not mean that they have to throw themselves out of the window to stop *in relation to the ground*. She means that the child should stop *with regard to the train.* Speed is a property of an object with respect to *another object*. It is a *relative* quantity.

The same is true for entropy. The entropy of A with regard to B counts the number of configurations of A that the *physical* interactions between A and B do not distinguish.

Clarifying this point, which frequently causes confusion, opens up a seductive solution to the mystery of the arrow of time.

The entropy *of the world* does not depend *only* on the configuration of the world; it also depends on the way in which we are blurring the world, and this depends on what the variables of the world are that *we* interact with. That is to say, on the variables with which our part of the world interacts.

The entropy of the world in the far past appears very low to us. But this might not reflect the exact state of the world: it might regard the subset of the world's variables with which *we*, as physical systems, have interacted. It is with respect to the dramatic blurring produced by our interactions with the world, caused by the small set of macroscopic variables in terms of which we describe the world, that the entropy of the universe was low.

This, which is a *fact*, opens up the possibility that it wasn't the universe that was in a very particular configuration in the past. Perhaps instead it is us, and our interactions with the universe, that are particular. We are the ones who determine a particular macroscopic description. The initial low entropy of the universe, and hence the arrow of time, may be more down to us than to the universe itself. This is the basic idea.